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1. Solution Overview

1.1 Executive Summary

OPOYO (Observe, Predict, On Your Own) is a passive, privacy-preserving eldercare monitoring system for seniors living independently. Current fall-detection systems rely on cameras and wearables, which raise privacy and user non-compliance concerns. OPOYO avoids these issues by using the reinforced concrete (RC) floor slab of an HDB flat to transmit vibrations. The system combines floor vibration data via piezoelectric sensors with ambient audio using local edge computing. This provides 24/7 fall detection, especially in privacy-sensitive areas like bathrooms, while minimising the false alarms that cause users to abandon monitoring devices.

1.2 Problem Statement: The "Long Lie" and Systemic Non-Compliance.

The medical emergency of a fall is rarely the physical impact itself; it is the subsequent inability to summon help.

- **The "Long Lie" Morbidity:** Literature establishes that roughly 47% of non-injured fallers cannot rise unaided (Tinetti et al., 1993). Remaining on the floor for over 60 minutes precipitates severe secondary complications, including dehydration, rhabdomyolysis, pressure injuries, and aspiration pneumonia, correlating with high six-month mortality rates (Fleming & Brayne, 2008; Wild et al., 1981).
- **Failure of Existing Infrastructure:** Singapore's existing safety net relies on wall-mounted Alert Alarm Systems (AAS). This assumes the senior is conscious, mobile, and in reach of the button. Wearables (pendants/watches) are frequently removed for charging, due to stigma, or during bathing, the exact window when fall risk peaks. Cameras are actively rejected in bedrooms and bathrooms.
- **The False Positive Failure Mode:** A passive system must be precise. A system that triggers false alarms for dropped objects is rapidly unplugged by users.

1.3 Target Demographic and Market Gap

The deployment target is Singaporeans aged 65 and above who live alone in HDB flats. The old-age support ratio in Singapore has fallen from 6.0 in 2014 to a projected 2.7 by 2030, making ambient human monitoring (living with relatives) increasingly unviable. This demographic requires continuous monitoring but actively resists visible surveillance. The stakeholders (adult children, Active Ageing Centres, and CareLine) require automated

alerting without the friction of manual check-ins. (For Jobs To Be Done analysis, refer to Appendix 6.2).

1.4 Solution Alignment

OPOYO addresses these systemic failures through four core architectural decisions:

1. **Contactless Sensing:** The user wears nothing and interacts with nothing. The sensor is a baseboard-mounted unit that monitors floor vibrations.
2. **Privacy-by-Design:** All digital signal processing (DSP) and machine learning inference executes locally on the edge microcontroller. No raw audio or vibration data is transmitted out of the home.
3. **Independent Sensor Veto:** By cross-referencing floor vibration with airborne sound, the system rejects environmental noise (e.g., dropped objects, television), eliminating the false-positive fatigue common in single-sensor systems.
4. **Automated Escalation:** Upon a confirmed fall, an automated API cascade messages primary caregivers and initiates a Twilio voice call to the senior, escalating to institutional care teams if unacknowledged.

2. Solution Features and Implementation Strategy

2.1 Physical Principles and Signal Characteristics

The core of the system relies on the physical differences between a human body and rigid objects, observed across two complementary media (solid concrete and air).

Structural Vibration

When a mass falls, it injects kinetic energy into the floor, generating structural Rayleigh waves. Modern HDB continuous reinforced concrete (RC) slabs transmit these low-frequency waves efficiently, with an attenuation factor of roughly 0.1-0.5 dB/m.

The structural vibration alone can differentiate a fall from a dropped object based on the temporal envelope:

- **Rigid Object:** A dropped book or pan creates a high-frequency, near-instantaneous impulse with a fast decay.
- **Human Body:** A human body is a soft, articulated, damped mass. A fall creates a lower-frequency, "smoother" wave envelope with a longer rise time and prolonged decay.

Airborne Sound

To reduce the variance in false positives (e.g., dropping a heavy, soft object like a gym bag that mimics a human impact), we cross-reference the structural data with airborne acoustics. Airborne sound is evaluated in the frequency domain. We apply a Fast Fourier Transform (FFT) to the raw microphone audio to separate the low-frequency "thud" of a fall from the high-frequency metallic ring of dropped objects or standard household noise (e.g., television).

By using two complementary sources, the probability of false positives is lowered. Because the error rates are not strictly statistically independent, the joint probability of a simultaneous false positive is modelled via conditional probability:

$$P(FP_{total}) = P(FP_{vibration} \cap FP_{sound}) = P(FP_{vibration}) \cdot P(FP_{sound} | FP_{vibration})$$

This architecture succeeds because the conditional probability remains exceptionally low for most deceptive household events.

2.2 Feature Breakdown

- **F1: Sensor Fusion:** Combines structural floor data with airborne acoustics. This lowers the false-positive rate, preventing user alarm fatigue and product abandonment.
- **F2: Edge Compute for Privacy:** 100% of signal processing occurs locally on the hardware node. Only basic status telemetry (e.g., fall status, confidence score) is transmitted, ensuring raw audio or vibration data never leaves the home.
- **F3: Escalation:** Upon confirmation of a fall classification, the system triggers an automated API escalation.
 - **t+0s:** A Telegram payload (room, timestamp) is pushed to the primary caregiver, alongside an automated Twilio voice call to the senior's phone.
 - **t+60s:** If the senior does not cancel the alarm or the caregiver does not acknowledge it, the system automatically escalates to a secondary contact or an institutional CareLine.

2.3 Technical and Strategic Implementation Approach

Hardware and Tech Stack

- **Production Hardware:** A sealed edge-node containing a piezoelectric contact transducer, a MEMS microphone, and a Raspberry Pi Zero 2 W microcontroller, powered by a 5V DC adapter and mounted directly to the floor skirting.
- **Hackathon Prototype:** A smartphone 3-axis MEMS accelerometer serves as the proxy (streamed via UDP with local timestamping to ensure synchronisation), networked to a laptop microphone to simulate the dual-sensor edge node.
- **Software Stack:** Python, NumPy, and SciPy for signal windowing, low-pass filtering, and FFT extraction. TensorFlow/Keras for kinematic neural network training; TensorFlow Hub for acoustic inference. FastAPI, Telegram Bot API, and Twilio API for telemetry.

Machine Learning Architecture

We use a deterministic, two-pipeline architecture to meet edge-compute latency constraints:

- **Kinematic Branch (1D-CNN):** Structural vibration arrays are segmented into 2-second windows and fed into a lightweight 1-Dimensional Convolutional Neural Network. The 1D convolutional kernels act as learnable Finite Impulse Response (FIR) filters, classifying the temporal decay signature of the impact.
- **Acoustic Branch (Zero-Shot Veto):** Audio data is fed into YAMNet, a pre-trained MobileNet architecture that classifies 521 environmental sounds. If the kinematic branch detects a fall but YAMNet simultaneously assigns high probabilities to non-human ambient noise (e.g., "Television," "Glass breaking," "Music"), the system vetoes the alert. Crucially, human vocalisation classes (e.g., "Speech," "Crying," "Groaning") explicitly bypass this veto logic to prevent false negatives if the senior calls for help following an impact.

2.4 Technical and Strategic Implementation Approach

Floor materials (e.g., ceramic tile vs vinyl) have highly variable mechanical damping coefficients. To prevent model degradation, the system executes a 60-second calibration upon deployment. The technician simulates negative impacts (dropping objects) and positive impacts (heavy heel-drops). Rather than attempting unstable on-device model retraining, the local microcontroller uses this localised dataset to establish a baseline amplitude gate and scale the input normalisation parameters, leaving the neural network weights frozen. This aligns the system with the room's specific acoustic signature without overfitting.

2.5 Validation Methodology

Due to the statistical rarity of human falls, baseline accuracy is a misleading metric. System robustness is validated strictly using the F1-Score and Precision-Recall AUC, calculated on a held-out adversarial dataset of deceptive events (slamming doors, dropping heavy objects). Our deployment threshold explicitly optimises for >90 recall while maintaining a projected False Positive Rate (measured in false alarms per sensor-month) low enough to ensure long-term hardware retention.

3. Solution Impact

3.1 Long-Term Impact

OPOYO's impact lies in consequence mitigation. The gerontological literature establishes a direct correlation between time on the floor and patient outcomes: roughly half of those who lie for over an hour die within six months (Wild et al., 1981). The mechanism of decline includes dehydration, pressure injuries, rhabdomyolysis, and aspiration pneumonia, which accumulate over time. Collapsing that exposure window from "up to an hour or more" to a deterministic, automated escalation ladder is the primary medical benefit of the system.

The second-order benefit is breaking the cascading consequences of a fall: prolonged hospitalisation, physical deconditioning, fear of falling, and subsequent activity restriction. Shortening the "long lie" reduces the likelihood that an acute event will necessitate permanent institutionalisation.

At a systemic level, this solution scales against degrading demographic constraints. Singapore's old-age support ratio fell from 6.0 in 2014 to 3.5 in 2024, and is projected to reach 2.7 by 2030. Active Ageing Centre (AAC) coordinators managing caseloads of 250+ seniors cannot rely on manual home visits to ensure safety. The system transitions AAC operations from scheduled wellness checks to targeted, data-driven interventions.

3.2 Unique Selling Points

OPOYO delivers specific utility to three distinct stakeholders (the senior, the family caregiver, and the institutional coordinator), mapping directly to the Jobs-to-Be-Done framework (Appendix 6.2).

Alternative	Where it fails this segment (1.2)	What OPOYO does instead
CareLine wall button	Requires the senior to be conscious, mobile, and willing to press the button, excluding severe falls.	Detects the physical impact and absence of recovery passively; requires zero action from the user.
Wearables (pendants)	Frequently removed due to stigma, charging requirements, or while showering/sleeping—when fall risk peaks.	Contactless monitoring. The senior wears nothing. Continuous coverage remains active in high-risk areas, such as the bathroom.
Cameras	Actively rejected in bathrooms and bedrooms due to privacy violations.	100% local edge processing. Only binary telemetry (timestamp, fall probability) is transmitted; raw audio or video never leaves the room.

- **Precision as Retention:** The independent sensor fusion (F1) solves the primary failure mode of passive monitors: abandonment due to false alarms. A system with a single-channel sensor solves detection; OPOYO solves retention by ensuring the device remains plugged in a year post-deployment.
- **Unified Hardware:** OPOYO satisfies the senior's requirement for privacy, the caregiver's requirement for immediate alerting, and the institution's requirement for triage data, all through a single hardware node without additional infrastructure.

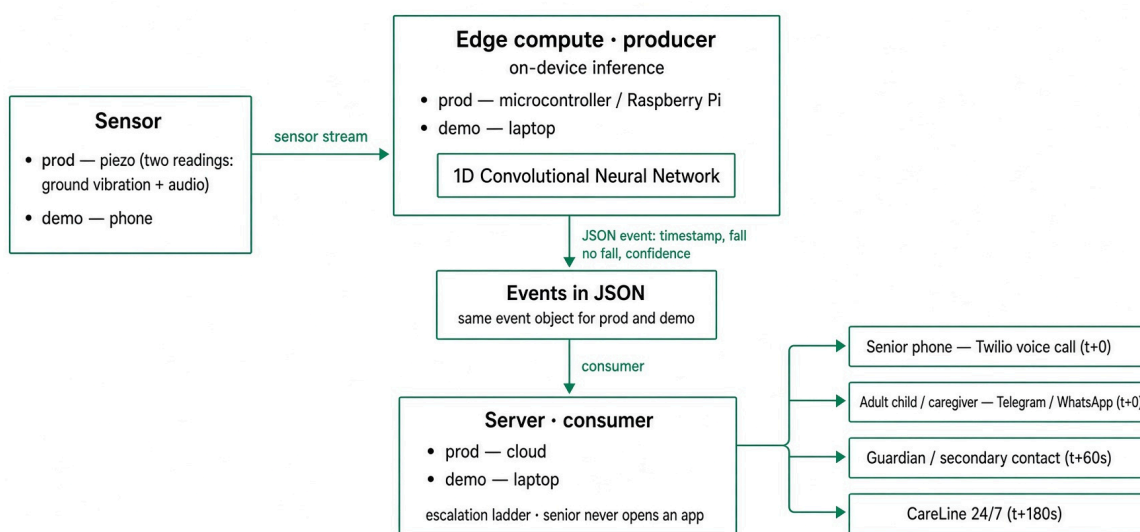
3.3 Key Performance Indicators

The core system metrics (precision, recall, and detection latency) serve as leading indicators for the following stakeholder outcomes:

Stakeholder	KPI	Target
Mdm Tan (user)	Median time from a confirmed fall to first human contact.	≤ 3 minutes (improving upon a documented status quo of 60+ minutes).
Mdm Tan (user)	Hardware retention rate.	$\geq 90\%$ of units are still active and unremoved at 12 months

Stakeholder	KPI	Target
Serene (buyer)	False alarms per sensor-month.	< 1 false alarm per month (the threshold required to prevent alarm fatigue and caregiver distrust).
Serene (buyer)	Caregiver reassurance score.	Self-reported reduction in anxiety-driven manual check-ins, tracked at 30/90/180 days.
AAC coordinator (institution)	Alert origination source.	Increasing percentage of falls reported via OPOYO API vs. delayed hospital admission notifications.
AAC coordinator (institution)	Recall rate at a fixed false-alarm ceiling.	>90% recall while strictly maintaining < 1 false alarm per sensor-month.

4. Solution Architecture



Same event object and escalation path for both prod and demo. Only the sensor and host hardware change.

5. Future Development and Scalability

While the current prototype serves as a reactive alarm, the hardware architecture supports future software-driven expansion into predictive monitoring and a third sensing channel.

5.1 Predictive Gait Analysis

Because the piezoelectric sensor continuously captures floor vibrations, it inherently records the resident's footsteps. Future software updates will introduce a continuous background DSP pipeline to extract step-time variability and stride asymmetry. Clinical literature indicates that increasing gait variability is an early, highly accurate indicator of an impending fall. By establishing a longitudinal baseline of the user's daily movement patterns, the system will shift from a reactive emergency trigger to a predictive healthcare tool, alerting AAC coordinators to a senior's mobility decline weeks before a physical fall occurs.

5.2 Third Sensor Integration (Wi-Fi Channel State Information)

To further reduce the false-positive rate, the production edge node will be updated to utilise Wi-Fi Channel State Information (CSI) sensing. Because the human body absorbs and reflects radio-frequency (RF) signals, human movement alters the multipath propagation of ambient Wi-Fi.

By applying custom firmware patches (e.g., Nexmon) to the edge node's Wi-Fi controller, the system can extract raw amplitude and phase shifts from Orthogonal Frequency-Division Multiplexing (OFDM) subcarriers. This allows the system to map gross human movement entirely through RF interference. By cross-referencing floor vibration, airborne sound, and RF CSI data, the system achieves a three-channel sensor fusion, maximising accuracy without requiring new hardware.

5.3 Deployment Roadmap and Institutional Integration

The scalability of the system relies on simple installation (one bonded sensor per isolated floor zone, such as a bathroom) and direct integration with Singapore's existing eldercare infrastructure.

- **Clinical Pilot:** Deployment in a controlled HDB rental block managed by a partner Active Ageing Centre (AAC). The objective is the empirical validation of the false-alarm rate (projected to be <1 per sensor-month in real-world conditions) and the verification of the API escalation latency.

- **CareLine Integration:** Current national infrastructure (e.g., the wireless Alert Alarm System) leaves a fatal coverage gap for seniors unable to physically press a wall button. This system will not compete with CareLine; it will integrate into it. By routing the automated escalation API into the existing Age Well SG and CareLine dispatch backend, the device serves as a passive, failsafe trigger for the nation's existing emergency response network

6. Appendix

6.1. Target Segment

Criterion	Assessment
Identifiable	Already recorded in AAC and HDB client registers. The segment is coherent (solo, mobile, monitoring-averse) and distinct from institutionalised or family-supported seniors.
Substantial	The number of seniors aged 65 and above living alone rose from 58,200 in 2018 to 78,100 in 2023, 87,200 in 2024, and 88,400 in 2025. Singapore is now a super-aged society, with about one in four citizens projected to be 65 or above by 2030.
Reachable	A dense institutional channel exists: 214 Active Ageing Centres, CareLine 24/7, and HDB rental-block programs under Age Well SG.
Responsive	This is the segment that has already refused or abandoned wearable alarms. A device demanding no behaviour change is one that has not been offered.
Profitable	Viable, but not through the senior. Revenue comes from the adult child and the institutional buyer.

Table 1 — Target segment assessed against the requirements for effective segmentation.

6.2. Jobs to Be Done

Who	Functional job	Emotional job	Social job
Mdm Tan (user)	When I fall alone, get help to me even if I cannot reach the phone.	Let me stop being afraid of my own bathroom at night.	Do not let anyone treat me as someone who needs watching.
Serene (buyer)	When I cannot reach my parent, tell me quickly if something is wrong.	Free me from having to leave work every time a call is missed.	Let me be a responsible caregiver without installing cameras in the house.

Who	Functional job	Emotional job	Social job
AAC coordinator	With 250+ clients and limited monthly visits, tell me who is declining.	Stop me from finding out about falls via delayed hospital notifications.	Let me justify and optimize where I spend my physical visits.

Table 2 — Jobs to be Done across user, buyer and institutional stakeholder.

Design Insight: The senior's social role—*not appearing to need watching*—is the exact friction point that undermines wearable pendants. Anything that must be worn, charged, or visibly displayed fails this requirement. This job can only be fulfilled by a system the user never physically interacts with, which dictates the floor-coupled sensor architecture and the automated phone call escalation.

6.3. Persona & Feature contribution to the identified problems.



Mdm Tan, 78
3-room Tampines, lives alone.
Walks to the market.
Took the CareLine pendant off after two weeks because it made her look old.
Falls risk is in the bathroom at night.
Job: get help without wearing anything or being on camera.



Serene, 46
Daughter in Punggol. Calls every evening.
Leaves work when mum misses two calls.
Pays if the thing is quiet and does not film the house.
Job: know if tonight is a fall, without another dashboard to babysit.



AAC coordinator
280 clients, 20 home visits a month.
Finds out about falls from the hospital.
Job: who declined this month, not another app login.

6.4. Principal Technical Risks and Mitigations

Risk	Consequence	Mitigation
Soft flooring (thick rug, foam-backed laminate) absorbs impact energy	Missed falls (False Negatives)	Installation protocol mandates hard-floor mounting; Section 3 Calibration measures achievable sensitivity at install and explicitly rejects uncoverable rooms. Deploy per-room nodes for decoupled zones.

Slab expansion joints reflect the wave	Reduced detection range in complex flat layouts	Coverage verified empirically per flat during the calibration walkthrough rather than assumed. A secondary node is deployed if a joint splits the unit.
Neighbour crosstalk through shared structural RC slabs	False positives triggered by adjacent flats	Shear walls and expansion joints between units heavily attenuate transmission. The calibration baseline and localised amplitude gating reject residual crosstalk.
Class imbalance in training data (falls are statistically rare)	Model collapses to "no fall" prediction	Class weighting during training, F1 and PR-AUC curve reporting, and an adversarial negative dataset. Standard "accuracy" is never utilised as a primary evaluation metric.
Domain shift from the reference dataset to the actual HDB floors	Degraded field accuracy and missed inferences	Few-shot on-device calibration (Section 3) scales input parameters; deployment utilises proprietary paired data collection. Limitations are stated transparently to the end-user.

6.5. Product MockUp

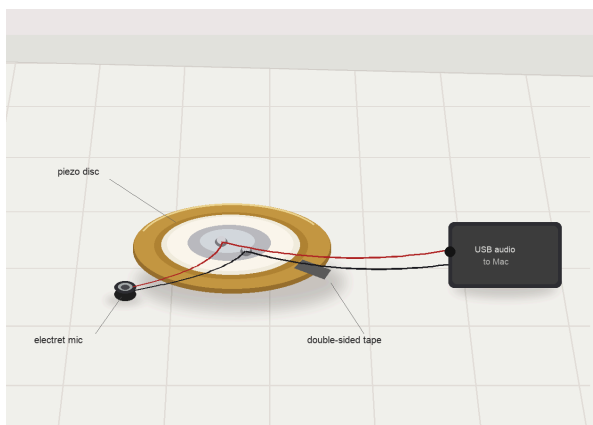


Figure A1. First hardware node (illustration). Brass piezo disc and electret mic coupled to the tile. Not the unit in this build. Demo uses phones on the



Figure A2. Final covered hardware node (illustration).

6.6. Wireframe

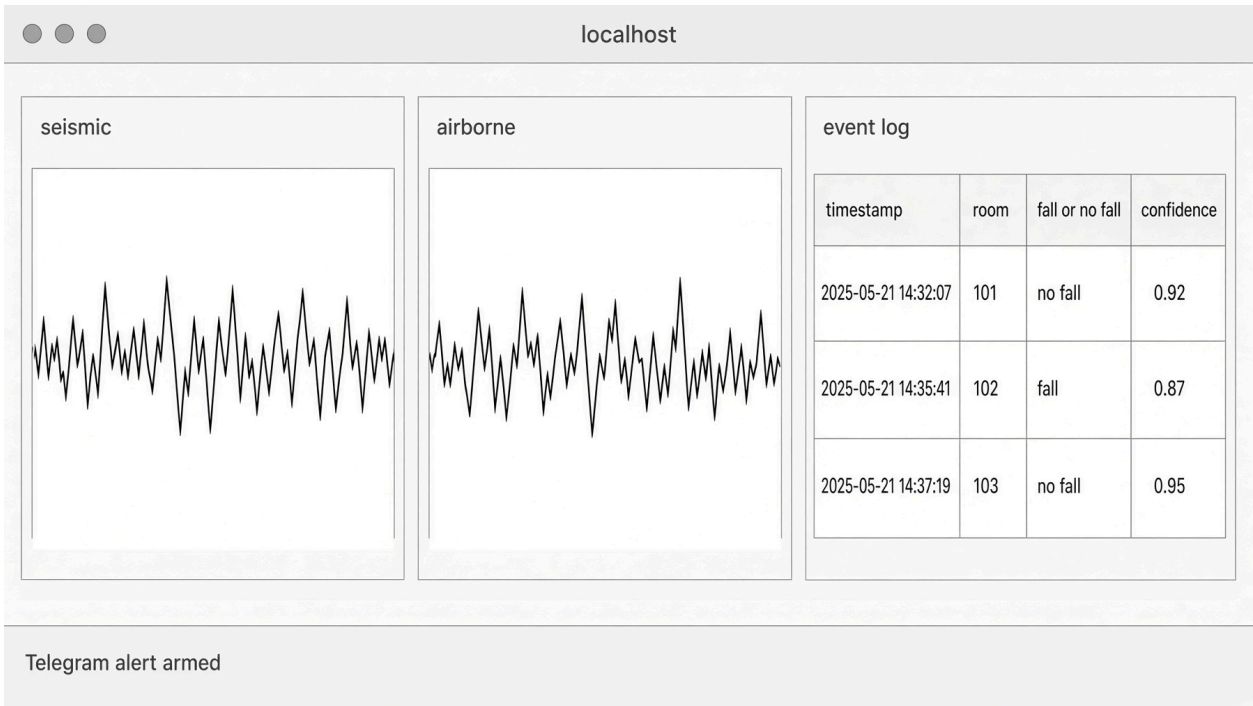


Figure A3. Desktop application, taking readings from the nodes

6.7. Demo Phone Client

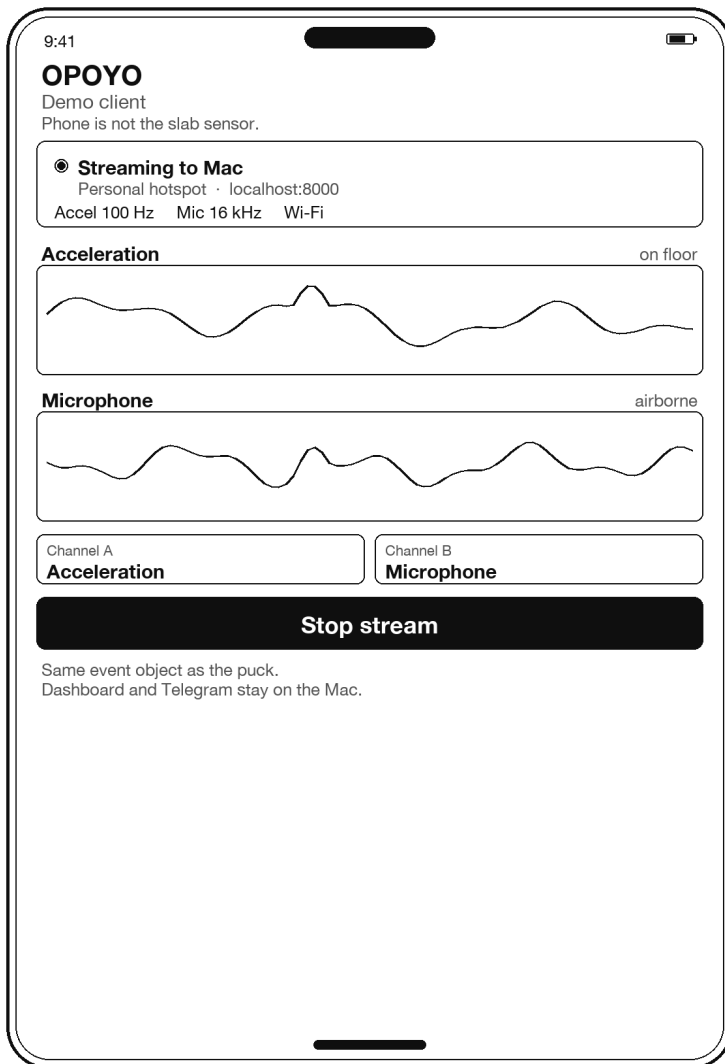


Figure A4. Streams acceleration and microphone to the Mac over a personal hotspot. The phone is not the slab sensor. Dashboard and Telegram stay on the Mac.

6.8. OPOYO Escalation Diagram

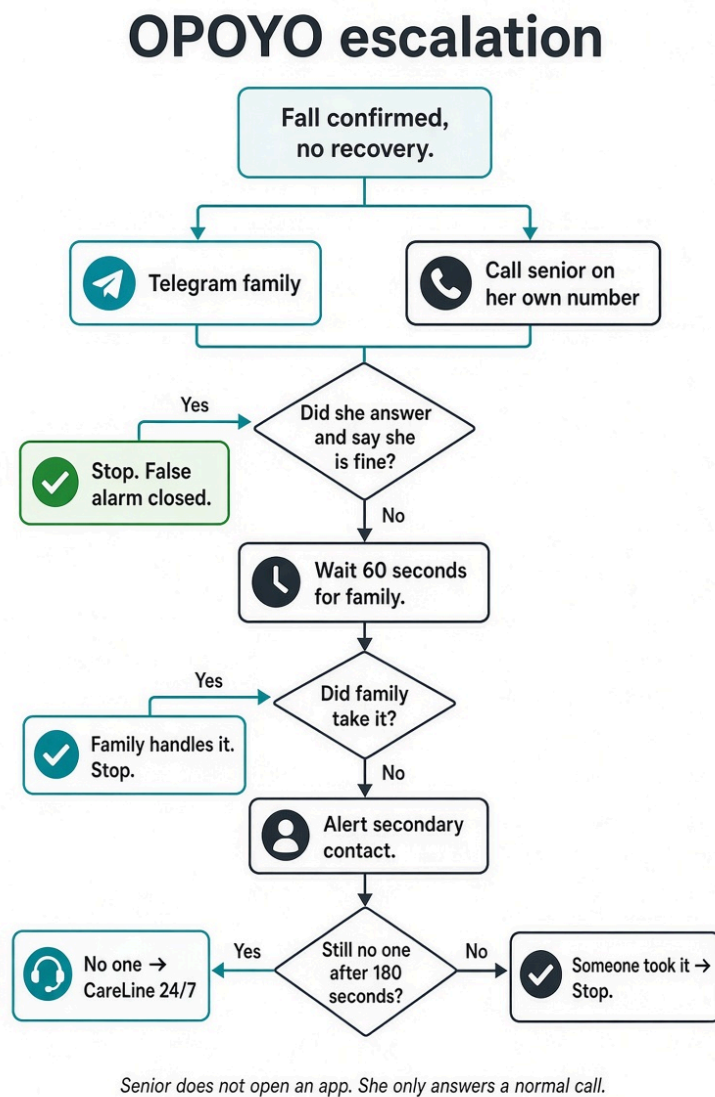


Figure A5: Timed ladder. Senior only answers a call. She does not open an app.

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